Group 6

CSE3006 – Computer Networks

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Topic
Advanced Encryption Standard (AES)



What is AES?

- **AES** stands for Advanced Encryption Standard.
- **Type:** a symmetric-key block cipher standardized by NIST.
- **Block size:** data is encrypted in fixed 128-bit blocks.
- **Key sizes:** 128-bit, 192-bit and 256-bit keys.
- **Origin:** based on the Rijndael cipher, selected by NIST in 2000.
- **Standard:** officially specified in FIPS 197 in 2001.

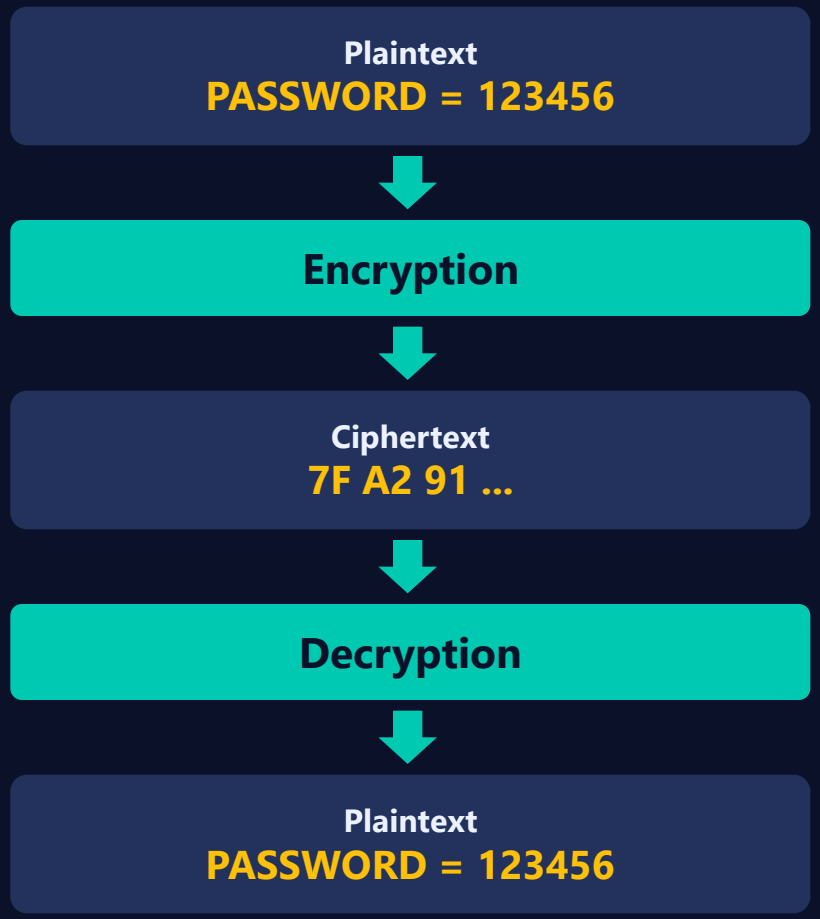


Key idea: Same secret key = Encryption + Decryption

MOTIVATION

Why Do We Need Encryption?

- Network data can be intercepted while in transit.
- Stored data can be accessed by unauthorized users.
- Encryption converts readable plaintext into ciphertext.
- Only a party with the correct key can recover the plaintext.
- It helps protect the confidentiality of sensitive information.
- Common targets: passwords, financial data, files and communications.



Important distinction: encryption primarily provides confidentiality. Authentication and integrity require additional mechanisms.

EVOLUTION OF ENCRYPTION STANDARDS

DES → 3DES → AES

DES – 1977

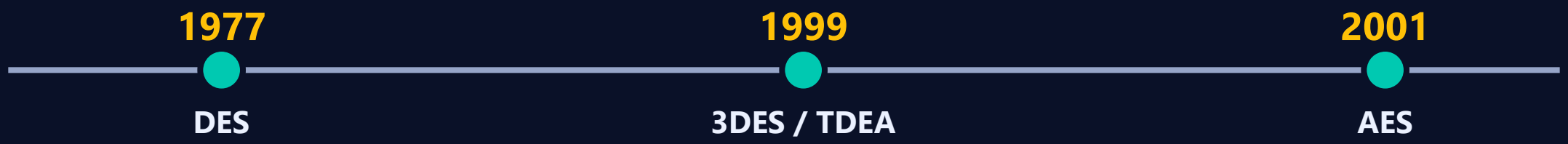
- Standardized as FIPS 46.
- Used a 56-bit effective key.
- Growing computing power made exhaustive key search practical.
- Withdrawn by NIST in 2005.

3DES / TDEA – 1999

- Applies DES-based operations multiple times.
- Introduced as a stronger transition from DES.
- Included in FIPS 46-3 (1999).

AES – 2001

- Based on the Rijndael cipher.
- Designed for stronger security and better performance and flexibility.
- Supports 128-, 192- and 256-bit keys.



Small key → stronger transition → modern symmetric standard

WHAT MAKES AES DIFFERENT

Why AES?

- **128-bit block size** regardless of key size.
- **Three key lengths:** 128, 192 and 256 bits.
- **Substitution–Permutation Network (SPN)** structure.
- Combines nonlinear substitution with byte-level permutation and mixing.
- Designed for efficient implementation in software and hardware.
- Substantially stronger security margins than legacy DES.



Design philosophy: Substitution + Permutation + Key Mixing → Confusion + Diffusion

STATE MATRIX

The AES State

4 × 4 State Matrix (16 bytes)

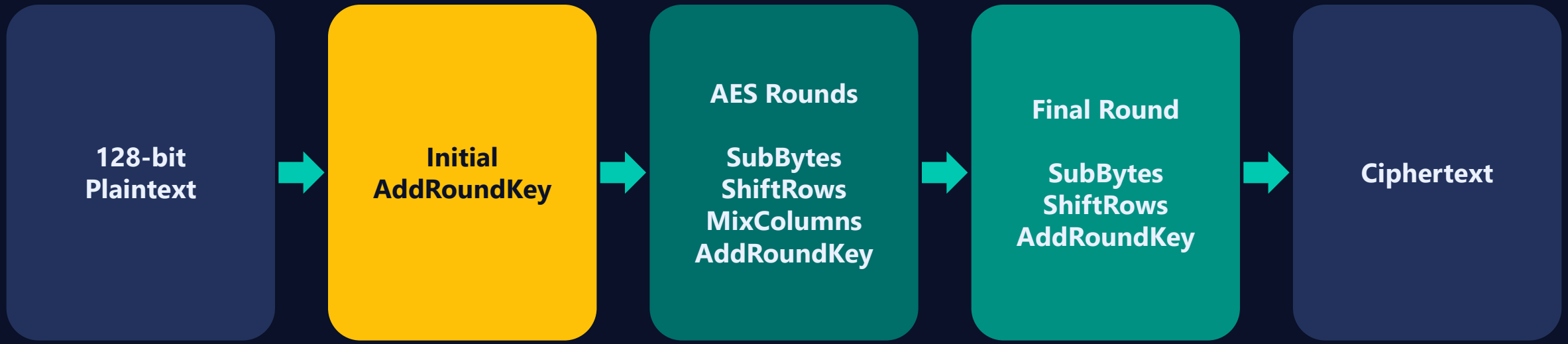
a0	a4	a8	a12
a1	a5	a9	a13
a2	a6	a10	a14
a3	a7	a11	a15

Input bytes fill the state column by column:
a0 a1 a2 a3 | a4 a5 a6 a7 | ...

- **One AES block** = 128 bits = 16 bytes.
- The bytes are arranged into a 4 × 4 state.
- All AES transformations operate on this state.
- The state is filled column-wise from the input block.
- The state changes after each transformation.

ENCRYPTION PROCESS

AES Encryption Pipeline



Important: the final round does not contain MixColumns.

This is one of the most important AES details to remember.

ENCRYPTION BEGINS

Step 1: Initial AddRoundKey

- AES begins by combining the plaintext state with the first round key.
- The operation uses XOR.
- Each state byte is XORed with the corresponding key byte.
- The secret key enters the encryption process immediately.
- No SubBytes, ShiftRows or MixColumns is performed before this step.



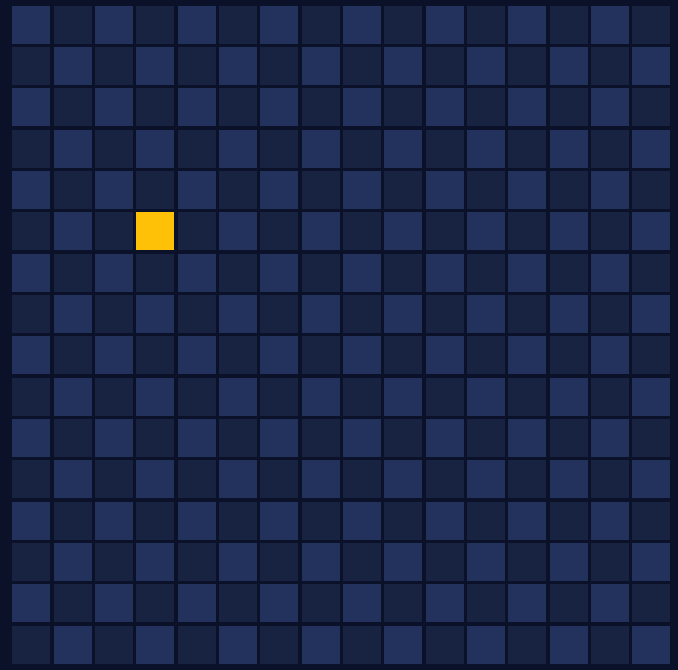

```
    10101100
XOR 01100101
    --------
    11001001
```



TRANSFORMATION 1

SubBytes – Nonlinear Substitution

- Each byte in the state is replaced using the AES S-box.
- The S-box provides nonlinearity.
- Every byte is substituted independently.
- The same S-box is used for all 16 state bytes.
- The inverse S-box is used during decryption.



Stylized 16 × 16 S-box
(row 5, column 3 highlighted)

Input Byte
53

→

AES S-box

→

Output Byte
ED

Why? Nonlinear substitution makes relationships between plaintext, ciphertext and key more difficult to exploit.

TRANSFORMATION 2

ShiftRows – Permuting the State

Before

A	B	C	D
E	F	G	H
I	J	K	L
M	N	O	P



After

A	B	C	D	no shift
F	G	H	E	left 1
K	L	I	J	left 2
P	M	N	O	left 3

Rules

- **Row 0:** no shift
- **Row 1:** shift left by 1 byte
- **Row 2:** shift left by 2 bytes
- **Row 3:** shift left by 3 bytes

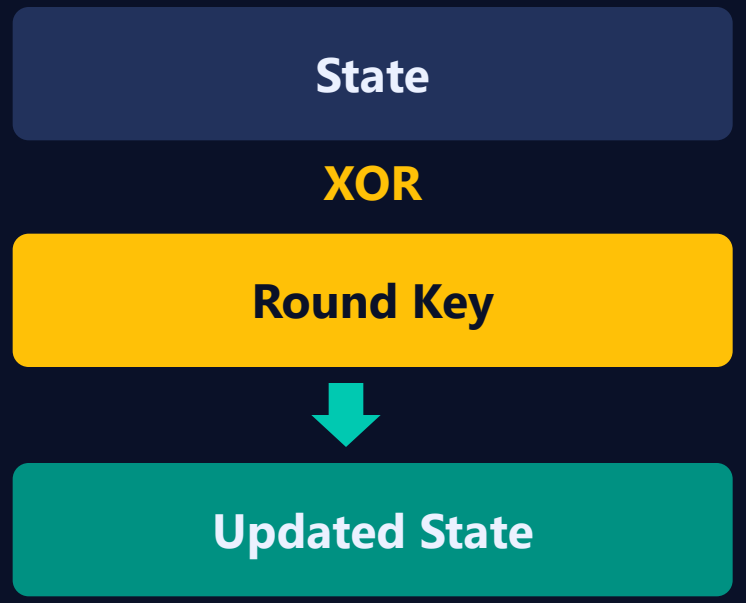
Purpose

- Rearranges bytes between columns.
- Helps spread the effect of substitutions across the state.
- Contributes to diffusion.

TRANSFORMATION 4

AddRoundKey – Key Mixing

- The current state is combined with a round key.
- The operation is a byte-wise XOR.
- Each AES round uses a different round key.
- Round keys are generated from the original key by the key schedule.
- This is where the secret key directly influences the round state.



Important property:

XOR is reversible

$$A \oplus K \oplus K = A$$

This property is useful during decryption.

ROUND STRUCTURE

Inside an AES Round

Standard round

SubBytes

↓

ShiftRows

↓

MixColumns

↓

AddRoundKey

(repeated Nr − 1 times)

Final round

SubBytes

↓

ShiftRows

↓

AddRoundKey

no MixColumns

Round count

AES-128 → 10 rounds

AES-192 → 12 rounds

AES-256 → 14 rounds

Key takeaway: MixColumns is intentionally omitted from the final round.

How Are Round Keys Generated?

- AES does not use the original key unchanged in every round.
- The original key is expanded into a sequence of round keys.
- This process is called Key Expansion (the Key Schedule).
- It uses operations such as RotWord, SubWord and Rcon.
- Each round receives its corresponding derived round key.
- Longer AES keys require more expanded key material.



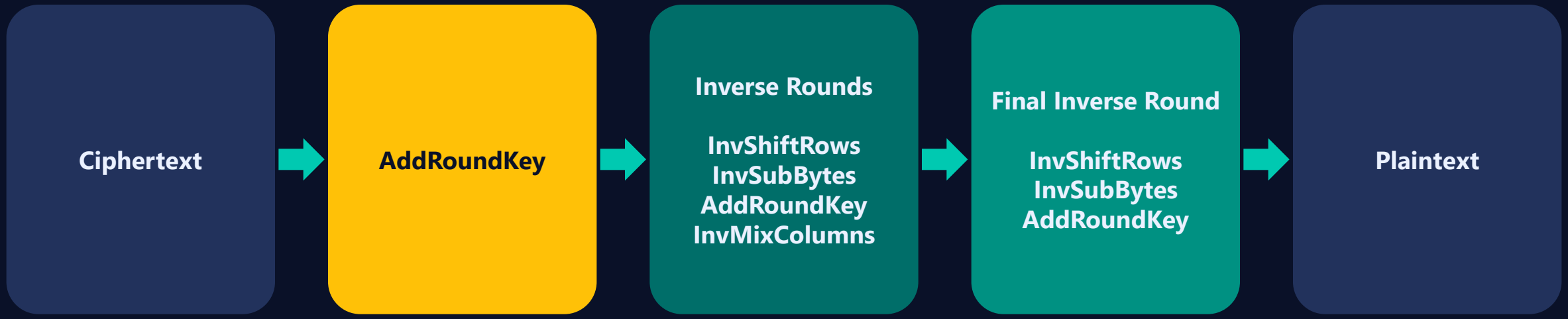
Technical note: the exact expansion differs between AES-128, AES-192 and AES-256.

REVERSING THE PROCESS

AES Decryption

Decryption reverses the encryption transformations using their inverse operations.

InvShiftRows | **InvSubBytes** | **AddRoundKey** | **InvMixColumns**



- Decryption uses the same secret key.
- Round keys are used in reverse order.
- In the final inverse round, InvMixColumns is omitted.

SECURITY

How Secure Is AES?

- AES has been extensively studied by the cryptographic community.
- No practical attack is known that recovers AES keys faster than brute-force search in the general case.
- Larger keys give larger key-search spaces: AES-256 > AES-128 (classically).
- Security also depends on correct implementation, key management and mode of operation.
- AES should not be described as 'unbreakable'.

Security hierarchy

AES-128
2^{128} possible keys

AES-192
2^{192} possible keys

AES-256
2^{256} possible keys

A strong cipher does not compensate for: weak passwords + leaked keys + implementation vulnerabilities

NIST continues to regard AES as an important symmetric primitive in post-quantum planning.

EVALUATION

Advantages vs Limitations

Advantages

- Strong security with 128/192/256-bit keys.
- Efficient in software and hardware.
- Widely standardized and extensively analyzed.
- Supports multiple modes of operation.
- Suitable for systems from servers to embedded devices.
- Hardware acceleration can provide high performance.

Limitations

- AES alone provides confidentiality, not complete security.
- Requires secure key management.
- Incorrect modes or nonce/IV handling can create serious vulnerabilities.
- AES does not inherently provide authentication or integrity.
- Security depends on correct implementation and system design.

Bottom line: Strong algorithm ≠ automatically secure system

SUMMARY

Conclusion

- AES is a symmetric-key block cipher standardized by NIST.
- It operates on 128-bit blocks with 128-, 192- or 256-bit keys.
- **Core transformations:** SubBytes, ShiftRows, MixColumns and AddRoundKey.
- AES uses 10, 12 or 14 rounds, depending on key size.
- The final encryption round omits MixColumns.
- AES remains a fundamental building block for modern data protection.
- Secure deployment also needs authenticated modes, proper key management and correct implementation.

AES provides the cryptographic engine – secure system design determines how safely that engine is used.

SOURCES

References

Primary Standards

[1] National Institute of Standards and Technology (NIST), FIPS 197: Advanced Encryption Standard (AES), Updated Edition, 2023. https://csrc.nist.gov/pubs/fips/197/final

[2] Morris Dworkin, NIST, SP 800-38D: Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC, 2007. https://csrc.nist.gov/pubs/sp/800/38/d/final

[3] NIST, Report on the Development of the Advanced Encryption Standard (AES), 2001.

Historical Standards

[4] NIST, FIPS 46-3: Data Encryption Standard (DES), 1999.

[5] NIST, FIPS 46: Data Encryption Standard (DES), 1977.

Textbooks

[6] William Stallings, Cryptography and Network Security: Principles and Practice, Pearson.

[7] Behrouz A. Forouzan, Data Communications and Networking, McGraw-Hill.